

10. (i) $\tan(x+y) - \sec(x+y) = C + x$. (ii) $ae^{-by} + be^{ax} = C$.

11. $(e^x + 2)\sec y = 3\sqrt{2}$.

12. (i) $y = Ce^{x/a}$;

(ii) $y^2 = 2ax + C$.

(iii) $r(C - \theta) = a$.

(iv) $r = a\theta + C$.

16.4. Homogeneous Equations.

If M and N of the equation $M dx + N dy = 0$ are both of the same degree in x and y and are homogeneous, the equation is said to be *homogeneous*. Such an equation can be put in the form

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right).$$

Every homogeneous equation of the above type can be easily solved by putting $y = vx$ where v is a function of x , and consequently

$$\frac{dy}{dx} = v + x\left(\frac{dv}{dx}\right), \text{ whereby it reduces to the form } v + x\left(\frac{dv}{dx}\right) = f(v),$$

i.e., $\frac{dx}{x} = \frac{dv}{f(v) - v}$ in which the variables are separable as shown below.

Ex. Solve $(x^2 + y^2) dx - 2xy dy = 0$.

The equation can be written as $\frac{dy}{dx} = \frac{x^2 + y^2}{2xy}$.

Putting $y = vx$, so that $\frac{dy}{dx} = v + x\frac{dv}{dx}$, we have

$$v + x\frac{dv}{dx} = \frac{x^2 + v^2x^2}{2vx^2} = \frac{1 + v^2}{2v}$$

$$\therefore x\frac{dv}{dx} = \frac{1 + v^2}{2v} - v = \frac{1 - v^2}{2v}$$

$$\therefore \frac{dx}{x} - \frac{2v}{1 - v^2} dv = 0.$$

$$\therefore \text{integrating, } \log x + \log(1 - v^2) = \log C.$$

$$\therefore x(1 - v^2) = C.$$

Re-substituting $\frac{y}{x}$ for v and simplifying, we get the solution

$$x^2 - y^2 = Cx.$$

16.5. A Special Form.

The equation of the form

$$\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2} \quad \left(\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \right) \quad \dots \quad (1)$$

can be easily solved by putting $x = x' + h$ and $y = y' + k$, where h and k are constants, so that $dx = dx'$ and $dy = dy'$ and choosing h, k in such a way that

$$\left. \begin{aligned} a_1h + b_1k + c_1 &= 0 \\ \text{and } a_2h + b_2k + c_2 &= 0 \end{aligned} \right\} \quad \dots \quad (2)$$

For, now the equation reduces to the form

$$\frac{dy'}{dx'} = \frac{a_1x' + b_1y'}{a_2x' + b_2y'},$$

which is homogeneous in x' and y' and hence solvable by the method of the previous article.

Note 1. The above method obviously fails, if $\frac{a_1}{a_2} = \frac{b_1}{b_2}$; for in this case h and k cannot be determined from equation (2).

Let the equation be

$$\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2} \quad \left(\frac{a_1}{a_2} = \frac{b_1}{b_2} \right) \quad \dots \quad (3)$$

$$\text{Let } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{1}{m}. \quad \therefore \quad a_2 = a_1m, \quad b_2 = b_1m,$$

where m is a non-zero constant.

Assuming this to be the case, let the common value of these ratios be denoted by $\frac{1}{m}$, so that $a_2 = a_1m$ and $b_2 = b_1m$.

The equation (3) becomes

$$\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{m(a_1x + b_1y) + c_2}.$$

Now, putting $a_1x + b_1y = v$, the variables can be easily separated and hence the equation can be solved. [See Ex. 2, below.]

Note 2. If in the equation (1), $a_2 = -b_1$, then the equation can be solved more easily by grouping the terms suitably.

[See Ex. 1(iv) of Examples XVI (C).]

16.6. Illustrative Examples.

Ex. 1. Solve $\frac{dy}{dx} = \frac{6x - 2y - 7}{2x + 3y - 6}$.

Putting $x = x' + h$, $y = y' + k$, so that $dx = dx'$, $dy = dy'$, we have

$$\frac{dy'}{dx'} = \frac{6x' - 2y' + 6h - 2k - 7}{2x' + 3y' + 2h + 3k - 6}.$$

Putting $6h - 2k - 7 = 0$ and $2h + 3k - 6 = 0$, and solving these two equations, we have $h = \frac{3}{2}$, $k = 1$.

$\therefore$ the equation becomes $\frac{dy'}{dx'} = \frac{6x' - 2y'}{2x' + 3y'}$.

Since the equation is now homogeneous, putting $y' = vx'$ and hence

$\frac{dy'}{dx'} = v + x' \frac{dv'}{dx'}$, and simplifying, the equation reduces to

$$\frac{dx'}{x'} = -\frac{1}{2} \frac{6v + 4}{3v^2 + 4v - 6} dv, \text{ which on integrating gives}$$

$$-\log Ax' = \frac{1}{2} \log (3v^2 + 4v - 6).$$

$$\therefore (Ax')^{-1} = (3v^2 + 4v - 6)^{\frac{1}{2}}.$$

Now, restoring the values of x' and v' , where $x' = x - \frac{3}{2}$ and

$v = \frac{y'}{x'} = 2 \cdot \frac{y - 1}{2x - 3}$, we get the solution in the form

$$3y^2 + 4xy - 6x^2 - 12y + 14x = C.$$

Ex. 2. Solve $\frac{dy}{dx} = \frac{6x - 2y - 7}{3x - y + 4}$.

Since here $\frac{a_1}{a_2} = \frac{b_1}{b_2}$,

$\therefore$ putting $3x - y = v$, we get

$3 - \frac{dy}{dx} = \frac{dv}{dx}$, and hence the given equation gives

$$\frac{dv}{dx} = 3 - \frac{2v - 7}{v + 4} = \frac{v + 19}{v + 4}.$$

$$\therefore dx = \frac{v + 4}{v + 19} dv = \left(1 - \frac{15}{v + 19}\right) dv.$$

$$\therefore x + C = v - 15 \log(v + 19).$$

On restoring the value of v , we get the solution in the form

$$2x - y - 15 \log(3x - y + 19) = C.$$

Ex. 3. Show that in an equation of the form

$$yf_1(xy)dx + xf_2(xy)dy = 0,$$

the variables can be separated by the substitution $xy = v$.

Since $xy = v$, $y = \frac{v}{x}$ and $d(xy) = dv$, i.e., $ydx + xdy = dv$

and $dy = \frac{xdv - vdx}{x^2}$, i.e., $xdy = dv - \frac{v}{x}dx$.

$$\therefore \frac{v}{x}f_1(v)dx + f_2(v)\left\{dv - \frac{v}{x}dx\right\} = 0.$$

$$\therefore \frac{f_2(v)dv}{v\{f_1(v) - f_2(v)\}} + \frac{dx}{x} = 0.$$

Thus, the variables are separated.

[See Ex. 14, 15, 16 of Examples XVI (B).]

We can as well form an equation in v and y , by taking $xy = v$,

$$x = \frac{v}{y} \text{ and } dx = \frac{ydv - vdy}{y^2}.$$

[For illustration see Alternative proof of Ex. 6 of Art. 16. 10.]

EXAMPLES XVI (B)

Solve (Ex. 1. - 15) :-

1. (i) $x + y \frac{dy}{dx} = 2y.$

~~(ii) $\frac{dx}{dy} + \frac{y}{x} = \frac{y^2}{x^2}.$~~

2. (i) $\frac{dy}{dx} = \frac{y(x-2y)}{x(x-3y)}.$

~~(ii) $\frac{dy}{dx} = \frac{x^2 y}{x^3 + y^3}.$~~

~~3. $(x^2 + y^2) dy = xy dx.$~~

~~4. (i) $\frac{dy}{dx} = \frac{x-y}{x+y}.$~~

~~(ii) $\frac{dy}{dx} = \frac{y(y+x)}{x(y-x)}.$~~

5. (i) $\left(3x \sinh \frac{y}{x} + 5y \cosh \frac{y}{x} \right) dx - 5x \cosh \frac{y}{x} dy = 0.$

(ii) $(1 + 3e^{\frac{x}{y}}) dx + 3e^{\frac{x}{y}} \left(1 - \frac{x}{y} \right) dy = 0.$

~~6. $(x^2 - 2xy) dy + (x^2 - 3xy + 2y^2) dx = 0.$~~

~~7. $y^2 dx + (x^2 + xy) dy = 0.$~~

~~8. (i) $\frac{dy}{dx} = \frac{y}{x} + \tan \frac{y}{x}.$~~

~~(ii) $\frac{dy}{dx} = \frac{3x+2y}{2x-3y}.$~~

9. $(6x - 5y + 4) dy + (y - 2x - 1) dx = 0.$

10. $(x - 3y + 4) dy + (7y - 5x) dx = 0.$

11. $(2x - 2y + 5) dy - (x - y + 3) dx = 0.$

12. $(x + y + 1) dx - (2x + 2y + 1) dy = 0.$

13. $y(2xy + 1) dx + x(1 + 2xy + x^2 y^2) dy = 0.$

14. $x^2 y^3 dx + 3x^2 y dy + 2y dx = 0.$

15. $(1 + xy \cos xy) dx + x^2 \cos xy dy = 0.$

16. Show that $(4x + 3y + 1) dx + (3x + 2y + 1) dy = 0$ represents hyperbolas having as asymptotes $x + y = 0$, $2x + y + 1 = 0$.**ANSWERS**

1. (i) $y = x + Ce^{x/(y-x)}.$

(ii) $2x - y = Cx^2 y.$

2. (i) $y^3 e^{\frac{x}{y}} = Cx^2.$

4. (i) $y^2 + 2xy - x^2 = C$

5. (i) $x^3 = C \sinh^5 \frac{y}{x}.$

6. $y = x \log(Cx^{-1}).$

8. (i) $x = C \sin \frac{y}{x}.$

9. $(5y - 2x - 3)^4 = C$

10. $(3y - 5x + 10)^2 = C$

11. $2y - x + C = \log(y)$

12. $6y - 3x = \log(3x - y)$

13. $2x^2 y^2 \log y - 4xy = C$

15. $xe^{\sin xy} = C.$

16.7. Exact

The differential equation is exact if the functions of x and y , M and N , such that $M dx + N dy$ is a perfect differential.

Now, we know that if $M dx + N dy$ is a perfect differential, then

$$M dx + N dy = 0 \text{ should be exact.}$$

The method of solution is as follows :

"First integrate M with respect to x to get the terms in $N dy$ already obtained. This will be the solution."

$$2. (i) y^3 e^{\frac{x}{y}} = C x^2. \quad (ii) y^3 = C e^{\frac{x^3}{y^3}}. \quad 3. \quad y = C e^{\frac{x^2}{2y^2}}.$$

$$4. (i) y^2 + 2xy - x^2 = C. \quad (ii) xy = C e^{\frac{y}{x}}.$$

$$5. (i) x^3 = C \sinh^5 \frac{y}{x}. \quad (ii) x + 3y e^{\frac{x}{y}} = C.$$

$$6. y = x \log (C x^{-1}). \quad 7. xy^2 = C(x + 2y).$$

$$8. (i) x = C \sin \frac{y}{x}. \quad (ii) 3 \log (x^2 + y^2) = 4 \tan^{-1} \frac{y}{x} + C.$$

$$9. (5y - 2x - 3)^4 = C(4y - 4x - 3).$$

$$10. (3y - 5x + 10)^2 = C(y - x + 1).$$

$$11. 2y - x + C = \log (x - y + 2).$$

$$12. 6y - 3x = \log (3x + 3y + 2) + C.$$

$$13. 2x^2 y^2 \log y - 4xy - 1 = Cx^2 y^2.$$

$$14. x(xy - 2)^3 = C(xy - 1)^3.$$

$$15. x e^{\sin xy} = C.$$

16.7. Exact Equation.

The differential equation $M dx + N dy = 0$, where both M and N are functions of x and y , is said to be exact when there is a function u of x, y , such that $M dx + N dy = du$, i.e., when $M dx + N dy$ becomes a perfect differential.

Now, we know from Differential Calculus that $M dx + N dy$ should be a perfect differential if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. Hence, the condition that